

11. Bayesian Networks: Representation

Ack: Berkeley AI course

Outline

- Independence
- Bayes Network Introduction

Probabilistic Models

- Models describe how (a portion of) the world works
- **Models are always simplifications**
 - May not account for every variable
 - May not account for all interactions between variables
 - “All models are wrong; but some are useful.”
 - George E. P. Box
- What do we do with probabilistic models?
 - We (or our agents) need to reason about unknown variables, given evidence
 - Example: explanation (diagnostic reasoning)
 - Example: prediction (causal reasoning)
 - Example: value of information

Independence

- Two variables are *independent* if:

$$\forall x, y : P(x, y) = P(x)P(y)$$

- This says that their joint distribution *factors* into a product two simpler distributions
- Another form:

$$\forall x, y : P(x|y) = P(x)$$

- We write:

$$X \perp\!\!\!\perp Y$$

- Independence is a simplifying *modeling assumption*
 - *Empirical* joint distributions: at best “close” to independent
 - What could we assume for {Weather, Traffic, Cavity, Toothache}?

Example: Independence

$P_1(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$P(T)$

T	P
hot	0.5
cold	0.5

$P_2(T, W)$

T	W	P
hot	sun	0.3
hot	rain	0.2
cold	sun	0.3
cold	rain	0.2

$P(W)$

W	P
sun	0.6
rain	0.4

Conditional Independence

- $P(\text{Toothache}, \text{Cavity}, \text{Catch})$
- If I have a cavity, the probability that the probe catches in it doesn't depend on whether I have a toothache:
 - $P(+\text{catch} \mid +\text{toothache}, +\text{cavity}) = P(+\text{catch} \mid +\text{cavity})$
- The same independence holds if I don't have a cavity:
 - $P(+\text{catch} \mid +\text{toothache}, -\text{cavity}) = P(+\text{catch} \mid -\text{cavity})$
- Catch is *conditionally independent* of Toothache given Cavity:
 - $P(\text{Catch} \mid \text{Toothache}, \text{Cavity}) = P(\text{Catch} \mid \text{Cavity})$
- **Equivalent statements:**
 - $P(\text{Toothache} \mid \text{Catch}, \text{Cavity}) = P(\text{Toothache} \mid \text{Cavity})$
 - $P(\text{Toothache}, \text{Catch} \mid \text{Cavity}) = P(\text{Toothache} \mid \text{Cavity}) P(\text{Catch} \mid \text{Cavity})$
 - One can be derived from the other easily

Conditional Independence

- Unconditional (absolute) independence very rare (why?)
- *Conditional independence* is our most basic and robust form of knowledge about uncertain environments.
- X is conditionally independent of Y given Z: $X \perp\!\!\!\perp Y | Z$

if and only if:

$$\forall x, y, z : P(x, y | z) = P(x | z)P(y | z)$$

or, equivalently, if and only if

$$\forall x, y, z : P(x | z, y) = P(x | z)$$

Conditional Independence

- What about this example:

- Traffic
- Umbrella
- Raining

$$T \perp\!\!\!\perp U \mid R$$

- What about this example:

- Fire
- Smoke
- Alarm

$$F \perp\!\!\!\perp A \mid S$$

Conditional Independence and the Chain Rule

- Chain rule: $P(X_1, X_2, \dots, X_n) = P(X_1)P(X_2|X_1)P(X_3|X_1, X_2) \dots$

- Trivial decomposition:

$$P(\text{Traffic}, \text{Rain}, \text{Umbrella}) = \\ P(\text{Rain})P(\text{Traffic}|\text{Rain})P(\text{Umbrella}|\text{Rain}, \text{Traffic})$$

- With assumption of conditional independence:

$$P(\text{Traffic}, \text{Rain}, \text{Umbrella}) = \\ P(\text{Rain})P(\text{Traffic}|\text{Rain})P(\text{Umbrella}|\text{Rain})$$

- Bayes' nets / graphical models help us express conditional independence assumptions

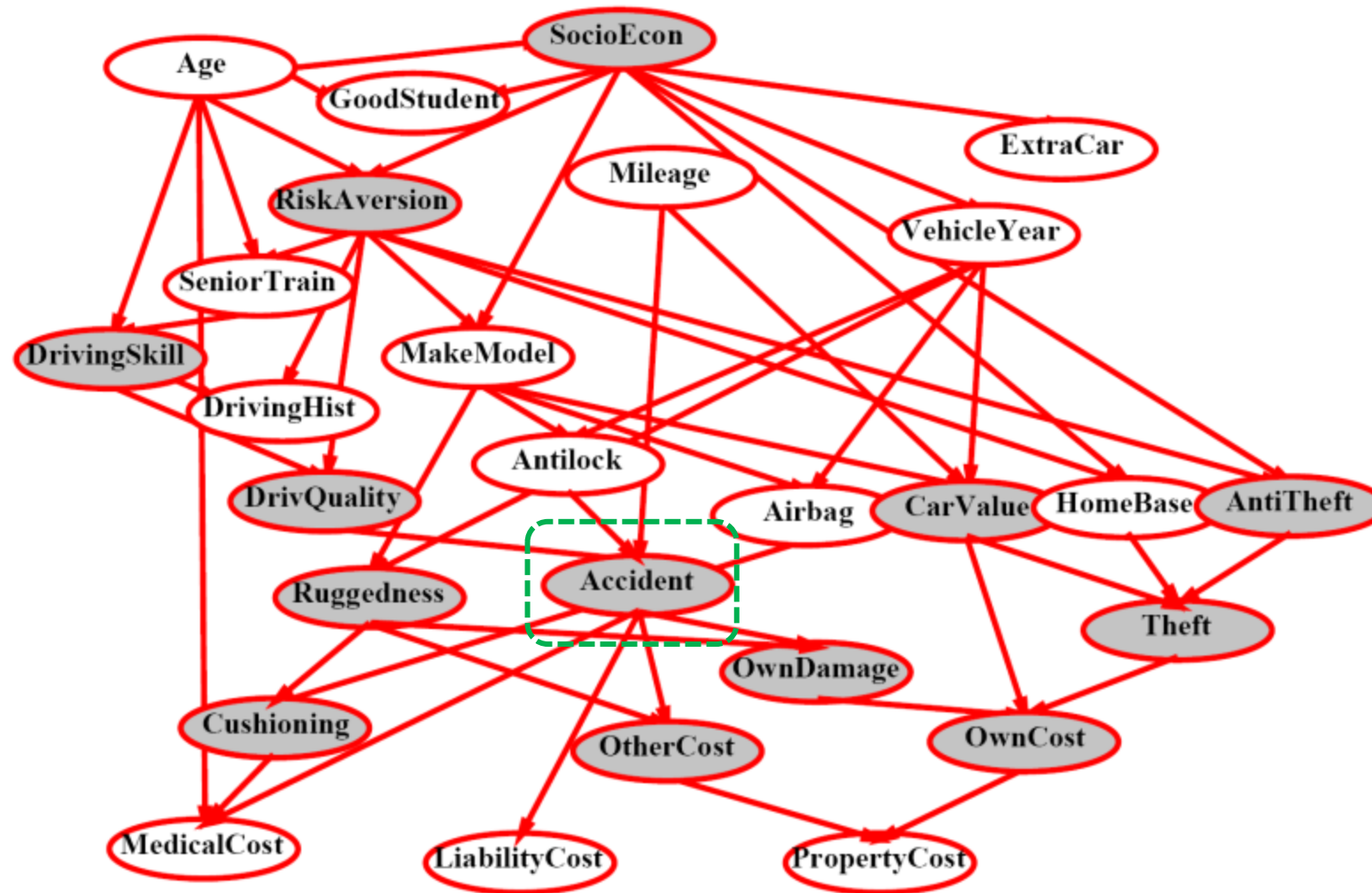
Outline

- Independence
- **Bayes Network Introduction**

Bayes' Nets: Big Picture

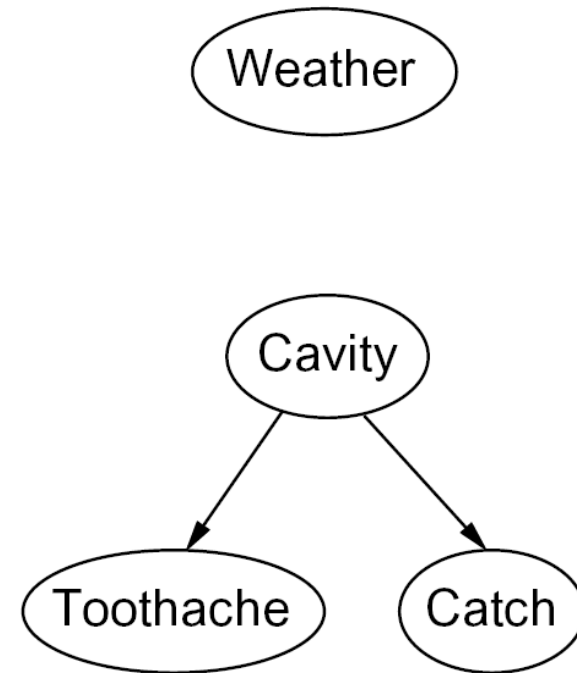
- Two problems with using full joint distribution tables as our probabilistic models:
 - Unless there are only a few variables, the joint is WAY too big to represent explicitly
 - Hard to learn (estimate) anything empirically about more than a few variables at a time
- **Bayes' nets:** a technique for describing complex joint distributions (models) using simple, local distributions (conditional probabilities)
 - More properly called **graphical models**
 - We describe how variables locally interact
 - Local interactions chain together to give global, indirect interactions

Example Bayes' Net: Insurance



Graphical Model Notation

- Nodes: variables (with domains)
 - Can be assigned (observed) or unassigned (unobserved)
- Arcs: interactions
 - Similar to CSP constraints
 - Indicate “direct influence” between variables
 - Formally: encode conditional independence (more later)
- For now: imagine that arrows mean direct causation (in general, they don't!)



Example: Coin Flips

- N independent coin flips



- No interactions between variables: **absolute independence**

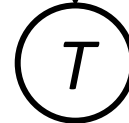
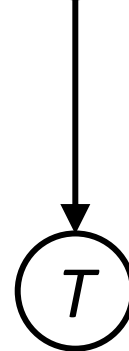
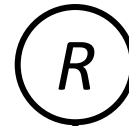
Example: Traffic

- Variables:
 - R: It rains
 - T: There is traffic

- Model 1: independence



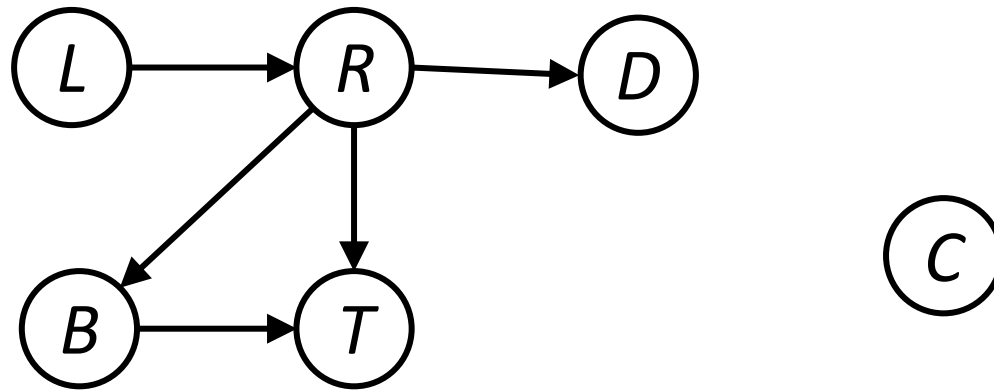
- Model 2: rain causes traffic



- Why is an agent using model 2 better?

Example: Traffic II

- Let's build a causal graphical model!
- Variables
 - T: Traffic
 - R: It rains
 - L: Low pressure
 - D: Roof drips
 - B: Ballgame
 - C: Cavity

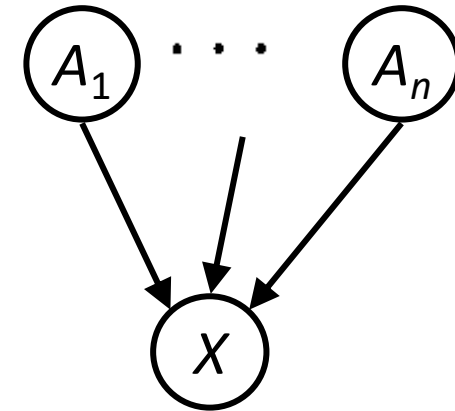


Bayes' Net Semantics

- A set of nodes, one per variable X
- A directed, acyclic graph
- A conditional distribution for each node
 - A collection of distributions over X , one for each combination of parents' values

$$P(X|a_1 \dots a_n)$$

- CPT: conditional probability table
- Description of a noisy “causal” process



$$P(X|A_1 \dots A_n)$$

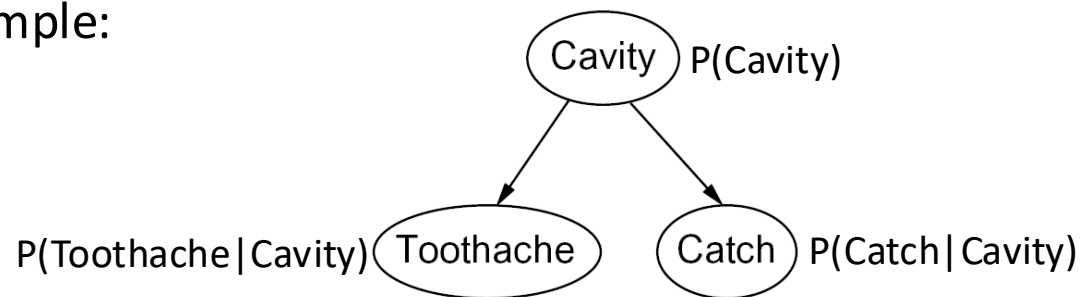
A Bayes net = Topology (graph) + Local Conditional Probabilities

Probabilities in BNs

- Bayes' nets **implicitly** encode joint distributions
 - As a product of local conditional distributions
 - To see what probability a BN gives to a full assignment, multiply all the relevant conditionals together:

$$P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i))$$

- Example:



$$P(+cavity, +catch, -toothache)$$

Probabilities in BNs

- Why are we guaranteed that setting

$$P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i))$$

results in a proper joint distribution?

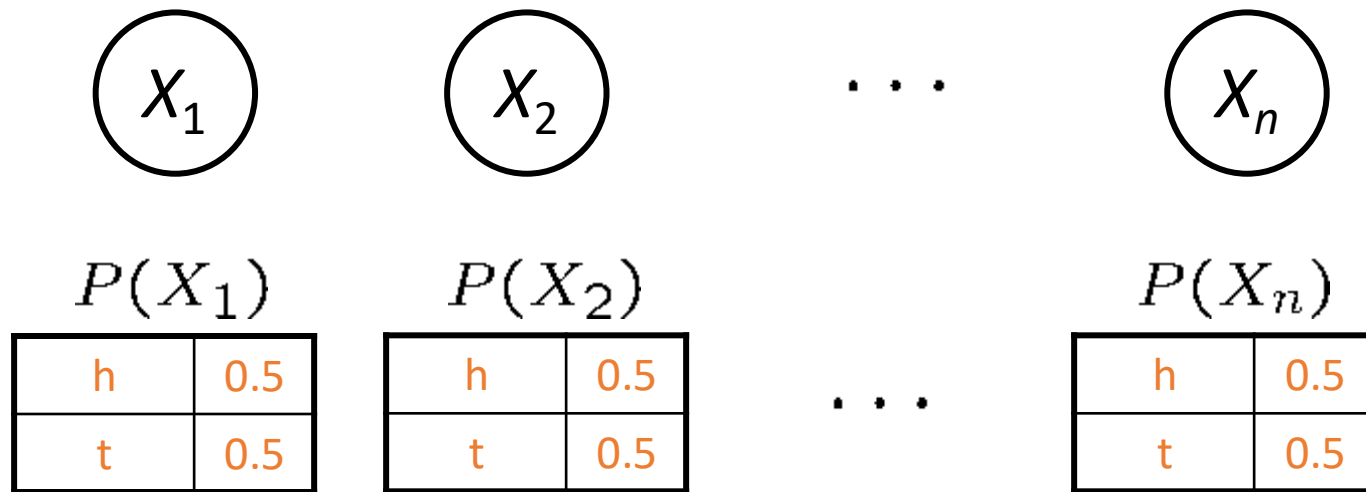
- Chain rule (valid for all distributions): $P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i | x_1 \dots x_{i-1})$
- Assume conditional independences: $P(x_i | x_1, \dots, x_{i-1}) = P(x_i | \text{parents}(X_i))$

→ Consequence:

$$P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i))$$

- Not every BN can represent every joint distribution
 - The topology enforces certain conditional independencies

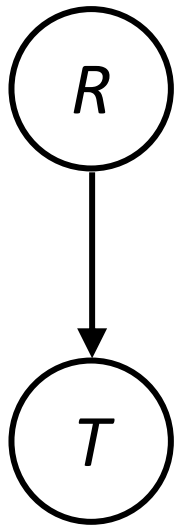
Example: Coin Flips



$$P(h, h, t, h) =$$

Only distributions whose variables are absolutely independent can be represented by a Bayes' net with no arcs.

Example: Traffic



$P(R)$

+r	1/4
-r	3/4

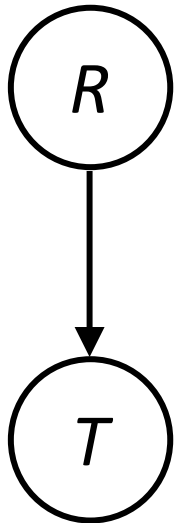
$P(T|R)$

+r	+t	3/4
+r	-t	1/4
-r	+t	1/2
-r	-t	1/2

$$P(+r, -t) = P(+r)P(-t|+r)$$

Example: Traffic

- Causal direction



$$P(R)$$

+r	1/4
-r	3/4

$$P(T|R)$$

+r	+t	3/4
	-t	1/4
-r	+t	1/2
	-t	1/2

$$P(T, R)$$

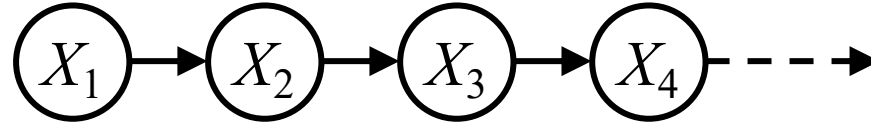
+r	+t	3/16
+r	-t	1/16
-r	+t	6/16
-r	-t	6/16

Reasoning over Time or Space

- Often, we want to **reason about a sequence** of observations
 - Speech recognition
 - Robot localization
 - User attention
 - Medical monitoring
- Need to introduce time (or space) into our models

Markov Models

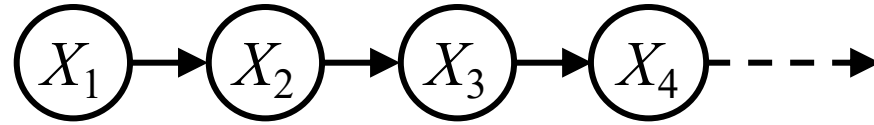
- Value of X at a given time is called the **state**



$$P(X_1) \quad P(X_t|X_{t-1})$$

- Parameters: called **transition probabilities** or dynamics, specify how the state evolves over time (also, initial state probabilities)
- Stationarity assumption: transition probabilities the same at all times

Markov Models



$$P(X_1) \quad P(X_t | X_{t-1})$$

- Basic conditional independence:
 - Past and future independent given the present
 - This is called the (first order) Markov property
- Note that the chain is just a (growable) BN
 - We can always use generic BN reasoning on it if we truncate the chain at a fixed length

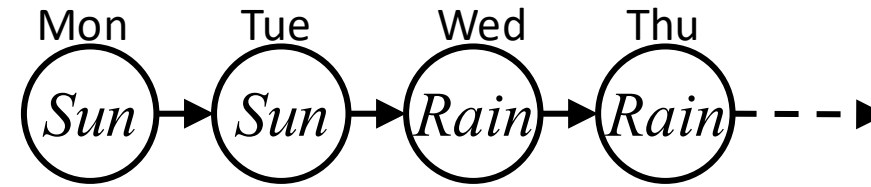
Example Markov Chain: Weather

- States: $X = \{\text{rain}, \text{sun}\}$

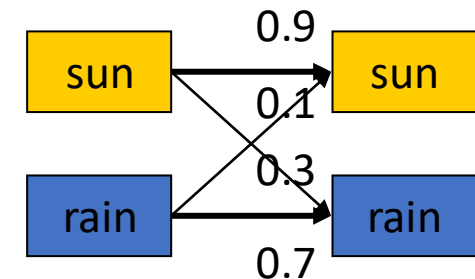
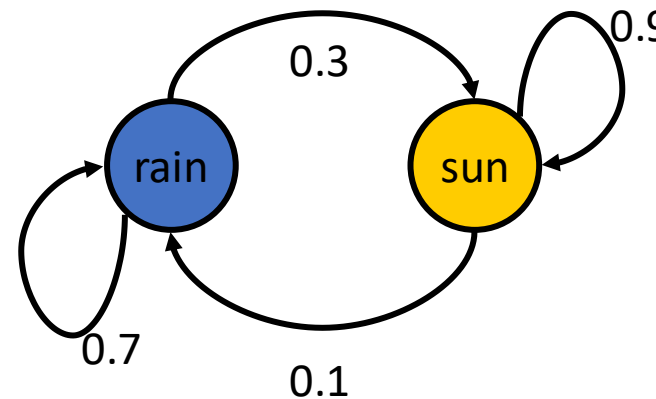
■ Initial distribution: 1.0 sun

■ CPT $P(X_t | X_{t-1})$:

X_{t-1}	X_t	$P(X_t X_{t-1})$
sun	sun	0.9
sun	rain	0.1
rain	sun	0.3
rain	rain	0.7



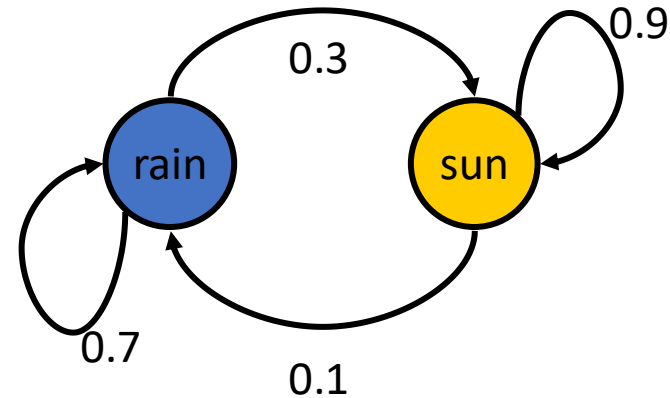
Two new ways of representing the same CPT*



*Same as MDP transition model, but no choice of action

Example Markov Chain: Weather

- Initial distribution: 1.0 sun

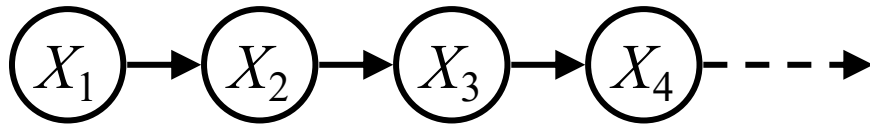


- What is the probability distribution after one step?

$$\begin{aligned} P(X_2 = \text{sun}) &= P(X_2 = \text{sun} | X_1 = \text{sun})P(X_1 = \text{sun}) + \\ &\quad P(X_2 = \text{sun} | X_1 = \text{rain})P(X_1 = \text{rain}) \\ &= 0.9 \cdot 1.0 + 0.3 \cdot 0.0 = 0.9 \end{aligned}$$

Mini-Forward Algorithm

- Question: What's $P(X)$ on some day t ?

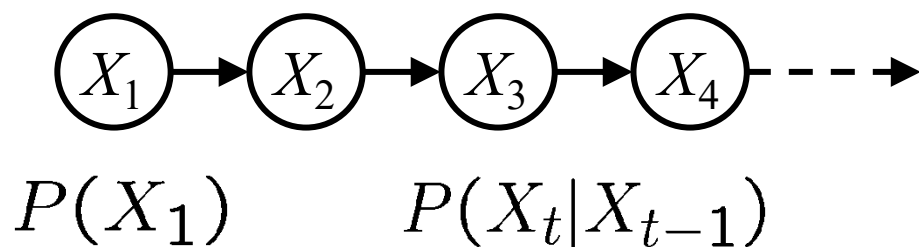


$$P(x_1) = \text{known}$$

$$\begin{aligned} P(x_t) &= \sum_{x_{t-1}} P(x_{t-1}, x_t) \\ &= \sum_{x_{t-1}} P(x_t \mid x_{t-1}) P(x_{t-1}) \end{aligned}$$

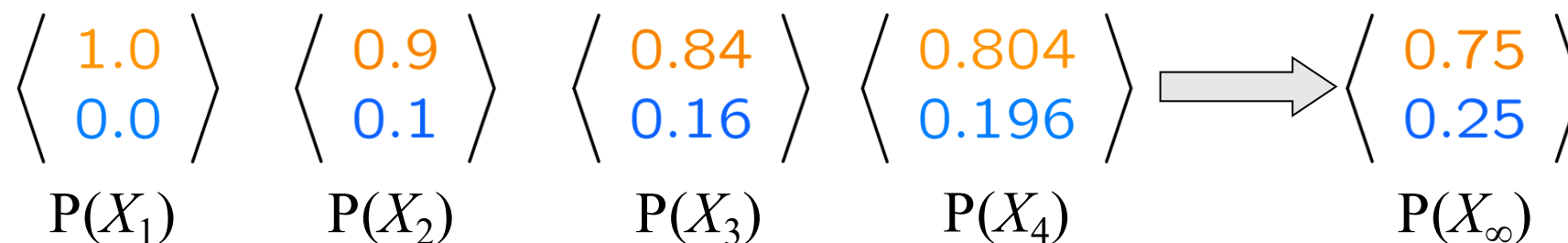
Forward simulation ←

Example Run of Mini-Forward Algorithm



X_{t-1}	X_t	$P(X_t X_{t-1})$
sun	sun	0.9
sun	rain	0.1
rain	sun	0.3
rain	rain	0.7

- From initial observation of sun



$$P(X_{\infty+1} = \text{sun}) = P(X_{\infty+1} = \text{sun} | X_\infty = \text{sun})P(X_\infty = \text{sun}) + P(X_{\infty+1} = \text{sun} | X_\infty = \text{rain})P(X_\infty = \text{rain})$$

$$= 0.9 * 0.75 + 0.3 * 0.25 = 0.75$$

Practice

- Given a Markov chain with two values for each state and the transition probabilities as shown below, please compute the stationary distribution.

